SelectNorm: A Disagreement-Factorized Analysis of Per-Input Normalization for Arabizi with Frozen Encoders

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Abstract

A classifier trained on Arabic-script text can be applied to Latin-script Arabizi directly or after transliteration, and the better choice varies from input to input. We ask whether, with frozen encoders and linear heads, this choice can be learned per input and transferred from Tunisian to Algerian and Moroccan Arabizi, and, if not, why. We use an elementary identity: for binary labels the expected benefit of normalizing is the probability that the two views disagree times the margin by which the normalized view wins when they do. The identity separates disagreement detection, conditional advantage estimation and a cost-aware decision, so each can be measured on its own. With four frozen encoders, normalizing improves macro-F1 by 2–19 points on Tunisian and Algerian sets, and on Moroccan text its sign depends on the encoder. Learned per-input routing does not consistently exceed strong baselines or the better single view. Against a direct-gain router every paired 95% interval lies within ± 0.7 points; against single views most intervals are inconclusive. Because routing must run the first pass before deciding, it costs more than either single view. The factorized analysis locates the limit for the features and estimators we test: disagreements are moderately predictable (AUC 0.62–0.74), but which view wins is weakly predictable from decision-point features (AUC at most 0.71, below 0.5 in two cases), also when a linear estimator is fitted on target labels. This is limited predictability under these choices, not unpredictability in principle. A bootstrap-classifier reference suggests that part of the oracle headroom is classifier variability. The usable signal is mostly set-level, which about a hundred labels recover.

1 Introduction

North African social-media text is written in Arabic script, in Latin-script Arabizi with digits standing for Arabic consonants (*3*=ع, *7*=ح), and with French segments inside the same sentence. A classifier trained on Arabic-script data can be applied to Arabizi in two ways: on the original string, or after transliterating it to Arabic script. Transliteration helps when it recovers words the classifier knows, and harms when it garbles French segments, negation, names or the spelling conventions of another dialect. The usual choice is made once for a whole dataset (Muller et al., 2021; Moosa et al., 2023; Artetxe et al., 2023), although the right choice varies from one input to the next.

This paper asks whether a frozen classifier can learn, without labels at inference time, *for which inputs* normalization pays off, and whether that decision transfers from Tunisian to Algerian and Moroccan Arabizi. Learning a per-input benefit predictor is not new: post-hoc deferral rules estimate the gain of a second stage (Jitkrittum et al., 2023), and loss predictors choose test-time transformations (Kim et al., 2020). What these methods do not address is how such a rule should be moved to a new dialect, where it is most likely to be wrong and where labels are scarce.

We start from an elementary identity. With binary labels, the expected benefit of normalizing an input equals the probability that the two views disagree, times the margin by which the normalized view wins when they do (Equation (3)). The first factor requires no labels; the second requires labels only on disagreements. The identity is not new in substance (it underlies disagreement-based methods such as query-by-committee), and we use it as an analysis device: it splits the routing problem into disagreement detection, conditional advantage estimation and a cost-aware decision, which can be estimated, adapted and *measured* separately. We ask whether per-input selection beats dataset-level choices and strong routers (RQ1), across corpora and dialects (RQ2); what it costs (RQ3); whether normalized predictions agree with predictions on human transliterations (RQ4); which sub-problem

limits it (RQ5); and how much a few target labels help (RQ6). All conclusions concern frozen encoders with linear heads. We contribute:

- a disagreement-factorized analysis of selective normalization: a cost-aware Bayes rule that separates sunk from avoidable inference cost, and estimators and adaptation regimes for each factor, built from standard tools (Sections 3 and 4);
- empirical evidence, across four frozen encoders and three Maghrebi varieties, on how far the *conditional advantage* of a normalized view, the factor that per-input routing hinges on, can be predicted from decision-point features with linear and boosted estimators (Section 6.3);
- a controlled, pre-specified evaluation with latency-based costs checked by end-to-end timing, paired tests with multiplicity correction, and interval estimates for the mechanism measurements (Sections 5 and 6);
- a conservative Arabizi normalizer with safeguards for French segments, negation and non-lexical cues; its intrinsic scores are development results, not an independent test (Table 6).

Normalization helps substantially on Tunisian and Algerian text, and its sign on Moroccan text depends on the encoder. SelectNorm stays within one point of the strongest router, but neither it nor any learned router consistently beats the better single view, and per-input routing is more expensive than either single view because the first pass is sunk. The analysis locates the limit in the conditional advantage: disagreements are moderately predictable, but the winning view is only weakly predictable from the features and estimators we test, even in domain. We report this rather than tune the method against the test sets.

2 Related Work

Deferral, cascades and routing. Selective classification abstains when confidence is low (Geifman and El-Yaniv, 2017, 2019); learning to defer trains a classifier and a deferral rule jointly (Mozannar and Sontag, 2020). Closest to our setting are post-hoc cascade rules: Jitkrittum et al. (2023) show that confidence-based deferral is optimal only when the second stage's errors are uniform and propose estimating the gain of deferring directly, and Gupta et al. (2024) learn deferral from token-level uncertainty. LLM routers learn which model to query from preference data (Ong et al., 2025) or from predicted weak–strong disagreement (Noorani et al., 2026). These methods choose between *models*; SelectNorm keeps one frozen model and chooses between two *views* of the input. Its estimator differs from direct gain prediction by factorizing the gain into a label-free disagreement propensity and a label-dependent conditional win rate (Equation (3)), which determines what can be adapted without labels and where labels are informative.

Selecting input transformations. Kim et al. (2020) predict, from the original image, the loss of each test-time augmentation and apply the best ones; our B4b baseline transfers this idea. Kang et al. (2026) train an LLM with reinforcement learning to decide whether to translate a query before reasoning. Translate-test is usually chosen per task rather than per input (Artetxe et al., 2023). None of these works studies the transfer of the selection rule across dialects.

Transliteration and dialect normalization. Transliterating unseen scripts can help multilingual encoders (Muller et al., 2021; Moosa et al., 2023), and Arabizi has been converted to Arabic script with rules, lexicons and sequence models (Darwish, 2014; Shazal et al., 2020). Dialect normalization maps dialectal text to a standard variety with rules and LLMs (Dimakis et al., 2025) or with dialect identification (Alhafni et al., 2024), and DialUp adapts models to dialect continua and dialect data to models (Bafna et al., 2025); Toughrai et al. (2025) model North African dialects from standard languages. These approaches transform all inputs. SelectNorm treats the transformation as an action whose benefit is estimated per input and can be withheld.

Using disagreement without labels. Agreement between predictors has been used to estimate accuracy on unlabeled data (Platanios et al., 2014; Garg et al., 2022), and querying labels where predictors disagree is the principle of query-by-committee (Seung et al., 1992). We use disagreement between two views of the same input to separate the label-free and label-dependent parts of a routing rule.

3 Problem Formulation

Setting. Let x be an input sentence with binary label $y \in \{0, 1\}$, T a deterministic normalizer that maps Latin-script Arabizi to Arabic script, and f_θ a frozen classifier with posterior $p_a = f_\theta(v_a)$ on view $v_0 = x$ or $v_1 = T(x)$ and decision $\hat{y}_a = \mathbf{1}[p_a >$

$1/2$]. A routing policy $g : \mathcal{Z} \to \{0, 1\}$ selects the view whose decision is returned, $\hat{y}(x) = \hat{y}_{g(z)}$, where $z = z(x)$ collects the features available at the *decision point*: after $f_\theta(x)$ and $T(x)$ have been computed and before $f_\theta(T(x))$ is run. Let $e_a = \mathbf{1}[\hat{y}_a \neq y]$. Measured inference costs, relative to the mean cost of $f_\theta(x)$, are $c_0(x)$ for the first pass, $c_T(x)$ for the normalizer and $c_1(x)$ for the second pass $f_\theta(T(x))$. Because z contains p_0 and normalizer metadata, $c_0 + c_T$ is *sunk* when g is decided; only c_1 is *avoidable*.

Objective and Bayes rule. For a trade-off $\lambda \geq 0$ we seek

$$\min_g \ \mathbb{E}\big[e_{g(z)} + \lambda \left(c_0 + c_T + c_1\, g(z)\right)\big]. \qquad (1)$$

The sunk term does not depend on g, so conditioning on z shows that the minimizer is

$$g^\star(z) = \mathbf{1}\big[B(z) > \lambda\, \bar{c}(z)\big], \qquad (2)$$

where $B(z) = \mathbb{E}[e_0 - e_1 \mid z]$ and $\bar{c}(z) = \mathbb{E}[c_1 \mid z]$, the expected *avoidable* cost. $B(z)$ is the expected *benefit* of normalizing. The sunk cost matters when routing is compared with other systems: a router that reads p_0 and m costs at least $c_0 + c_T$, whereas never normalizing costs c_0 and always normalizing costs $c_T + c_1$. Per-input routing of this kind can therefore save compute only relative to running both views, never relative to a single view. It is the quantity estimated by post-hoc deferral rules (Jitkrittum et al., 2023) and, for input transformations, by loss predictors (Kim et al., 2020); estimating it directly is therefore not new.

Disagreement factorization (elementary). Let $D = \mathbf{1}[\hat{y}_0 \neq \hat{y}_1]$ and $W = \mathbf{1}[\hat{y}_1 = y]$. Because $e_0 = e_1$ whenever $D = 0$, and because with binary labels exactly one view is correct whenever $D = 1$,

$$\begin{aligned} B(z) &= \pi(z)\,\big(2\rho(z) - 1\big), \qquad (3)\\ \pi(z) &= P(D{=}1 \mid z),\\ \rho(z) &= P(W{=}1 \mid D{=}1, z). \end{aligned}$$

(With K classes, $B = \pi\,(\rho_1 - \rho_0)$ with ρ_a the probability that view a is right given a disagreement.) The identity is elementary and is implicit in disagreement-based methods such as query-by-committee (Seung et al., 1992); we do not claim it as a theoretical contribution. We use it as an *analysis device*: the two factors have different data requirements and can be measured separately: π depends on the two views' predictions only, so it can be estimated on *unlabeled* text once both views are run offline; ρ needs labels, but only on disagreements, since labels of agreeing inputs carry no information about B.

Three sub-problems. Equations (2) and (3) split selective normalization into *disagreement detection* (π), *conditional advantage estimation* (ρ) and a *cost-aware decision* ($\lambda\bar{c}$). Both estimators are fitted by minimizing the log-loss, a proper scoring rule, on the policy split: $\hat{\pi}$ on all inputs with target D, $\hat{\rho}$ on disagreements with target W. The plug-in rule is $\hat{g} = \mathbf{1}[\hat{\pi}(2\hat{\rho} - 1) > \lambda\bar{c}]$. Two consequences follow directly.

(i) At zero cost the rule ignores π. For $\lambda = 0$ and $\hat{\pi} > 0$, $\hat{g} = \mathbf{1}[\hat{\rho} > 1/2]$: the decision depends only on the conditional advantage, so re-estimating π can change decisions only at $\lambda > 0$.

(ii) Errors of each factor are weighted by the other.

Proposition 1 (Plug-in regret)**.** *Let $R(g)$ be the risk in Equation (1). Then*

$$R(\hat{g}) - R(g^\star) \leq \mathbb{E}\big[\,|2\rho - 1|\,|\hat{\pi} - \pi| + 2\hat{\pi}\,|\hat{\rho} - \rho|\,\big].$$

The usual plug-in argument gives $R(\hat{g}) - R(g^\star) \leq \mathbb{E}|\hat{B} - B|$, and $\hat{B} - B = (\hat{\pi} - \pi)(2\rho - 1) + 2\hat{\pi}(\hat{\rho} - \rho)$ (Appendix A). Errors in ρ therefore cost most where disagreements are frequent. Errors in π cost little when neither view dominates ($\rho \approx 1/2$), which is also where any rule gains little. The bound is the standard plug-in argument applied to Equation (3) and is not a new learning-theoretic result; we use it only to motivate measuring the two estimation errors separately (Section 6.3).

Assumptions and limits. Exactness of Equation (3) requires binary labels and the 0-1 loss; macro-F1, our reported metric, is not a sum of per-input losses, so the rule is accuracy-optimal and only approximately macro-F1-optimal. Transfer of $\hat{\rho}$ to a new domain Q assumes $P_Q(W \mid D{=}1, z) = P(W \mid D{=}1, z)$; this conditional invariance is the assumption we test.

Adaptation to a new dialect. A router trained on Tunisian data meets a target distribution Q (another corpus or dialect). Re-estimating π_Q on unlabeled target text is label-free; whether this suffices depends on the stability of ρ across domains, which we test rather than assume (§6). When ρ shifts, we model $\rho_Q(z) = \sigma\big(\text{logit}\,\rho(z) + b\big)$ and estimate the scalar b from k labelled target disagreements.

Proposition 2 (Label efficiency of disagreement-targeted labelling)**.** *Under the model above, with π_Q known, the Fisher information about b carried by k target labels drawn at random is $k\,P_Q(D{=}1)\,\mathbb{E}_Q[\rho_Q(1-\rho_Q) \mid D{=}1]$, and by k labels drawn among disagreements it is $k\,\mathbb{E}_Q[\rho_Q(1-\rho_Q) \mid D{=}1]$. Targeted labelling is therefore $1/P_Q(D{=}1)$ times more informative.*

The proof (Appendix A) is a one-line Fisher-information calculation that uses only that an agreeing example's likelihood does not depend on b; it concerns the efficiency of estimating b, not the accuracy of the resulting router, which Section 6 tests. Labelling where predictors disagree is the idea of query-by-committee (Seung et al., 1992); here the target of estimation is a routing rule, and Equation (3) gives the exact reason why only disagreements matter.

Information access. The policy never reads p_1 at decision time; π and ρ are *predicted* from z. Training uses p_1 on the policy split only, and adaptation uses p_1 on unlabeled target text that is disjoint from the evaluated fold. The oracle $g^{\text{or}} = \mathbf{1}[e_0{=}1 \wedge e_1{=}0]$ reads labels and is reported only as an upper reference: ties and joint errors keep the original view at no cost.

4 SelectNorm

Module A: conservative normalizer. T processes Latin-script tokens and leaves Arabic-script tokens, digits-only tokens, emoji and punctuation unchanged (Algorithm 1). A token is first looked up in a lexicon of Arabizi→CODA pairs built from 80% of TArC (Gugliotta and Dinarelli, 2022); unseen tokens fall back to Tunisian character rules (*3*→ع, *7*→ح, *9*→ق, *5*→خ, *ch*→ش, word-initial vowels to alef, medial short vowels dropped). Three safeguards define T_{cons}: (i) *protection* of French and English tokens, decided from word frequencies against TArC evidence, with short or function-like tokens (*la*, *ma*, *el*) read in the language of their sentence context; (ii) *negation*: the Tunisian circumfix *ma…ch* is split as ما + stem + ش and frequent negators (*mouch*, *mech*) are mapped by a fixed list, so that the negation particle survives; (iii) *non-lexical cues*: laughter (*hhhh*, *mdr*) is mapped to هههه rather than transliterated letter by letter. Two weaker strategies serve as controls: T_{rule} (rules only) and T_{lex} (lexicon and rules, no safeguards). Every call returns token-level reliability metadata (lexicon hits, rule fallbacks, ambiguous characters, protected, negation and laughter tokens), which the router may read because it is computed before the second pass.

Algorithm 1 Conservative normalization T_{cons}

Require: sentence x, lexicon $\mathcal{L}$, rules R, frequency lists F
Ensure: $T(x)$, metadata m
$\tau \leftarrow$ tokens of x; $\kappa \leftarrow$ share of foreign tokens in τ by F vs. $\mathcal{L}$
for each Latin-script token $t \in \tau$ **do**
  if t is laughter **then** emit هههه
  else if t is foreign (short: iff $\kappa > 1/2$) **then** keep t
  else if t is a listed negator **then** emit its fixed form
  else if $t \in \mathcal{L}$ **then** emit $\mathcal{L}(t)$
  else if t matches *ma…ch* **then** emit ما R(stem)ش
  else emit $R(t)$ and count ambiguous characters
  update m with the branch taken
return joined output, m

Module B: benefit estimation. The decision-point features z form three groups: *conf* (maximum probability, margin and entropy of the temperature-scaled p_0), *script* (length, Latin share, digits used as letters, emoji and upper-case rates), and *trel* (the normalizer metadata as rates per Latin token, and the share of changed tokens). $\hat{\pi}$ and $\hat{\rho}$ are logistic regressions on standardized z (gradient boosting as ablation); $\hat{\rho}$ is fitted on the disagreements of the policy split only (Algorithm 2).

Module C: cost-aware routing. At inference (Algorithm 3) the classifier runs once on x, the normalizer runs (both costs are sunk at the decision point), and $\hat{B}(z) = \hat{\pi}(z)(2\hat{\rho}(z) - 1)$ is compared with a threshold: either $\lambda\bar{c}$, the Bayes threshold of Equation (2), or, as for every baseline, a threshold tuned for each λ on the policy development split. No threshold is tuned on target data. The second pass runs only when $g = 1$; its cost $\bar{c}$ is the only avoidable cost, and it is the only cost that enters the threshold. Reported costs count everything a system runs (Section 5).

Complexity. Let $C_f(n)$ be the cost of one encoder pass on n subword tokens. T is $O(n)$ with dictionary lookups; $\hat{\pi}, \hat{\rho}$ cost $O(|z|)$. Expected inference cost is $C_f + C_T + O(|z|) + \mathbb{E}[g]\,C_f$, of which only the last term is avoidable; training the router costs one pass of each view over the policy split and $O(N|z|)$ per logistic fit. Adaptation adds one offline pass of each view over unlabeled target text and k annotations.

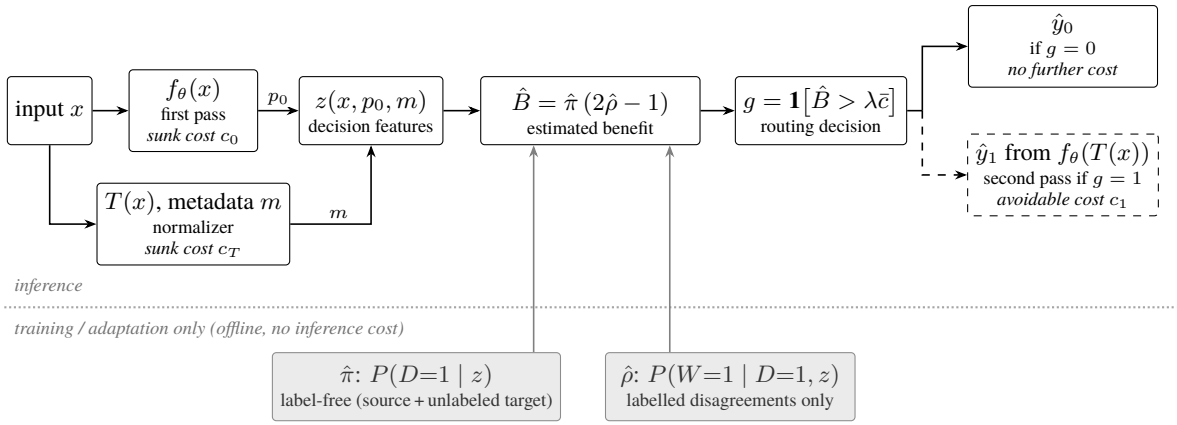


Figure 1: SelectNorm. Top lane, inference: $f_\theta(x)$ and $T(x)$ always run, so their cost is sunk when g is decided; the second pass $f_\theta(T(x))$ (dashed) is the only avoidable cost and runs only when the estimated benefit exceeds $\lambda\bar{c}$. Bottom lane, grey: estimators fitted offline for training or adaptation, with no inference cost beyond evaluating $\hat{\pi}, \hat{\rho}$.

Algorithm 2 Training (labels from the policy split only)

Require: frozen f, normalizer T, policy split $\mathcal{P} = \{(x_i, y_i)\}$; optional unlabeled target set $\mathcal{U}$; optional k labelled target disagreements $\mathcal{K}$
Ensure: $\hat{\pi}$, $\hat{\rho}$ (and shift $\hat{b}$)
for $x \in \mathcal{P} \cup \mathcal{U}$ **do** compute p_0, p_1, z, D
$\hat{\rho} \leftarrow$ fit $P(W{=}1 \mid z)$ on $\{i \in \mathcal{P} : D_i = 1\}$ ▷ needs y
$\hat{\pi} \leftarrow$ fit $P(D{=}1 \mid z)$ on $\mathcal{P}$, or on $\mathcal{P} \cup \mathcal{U}$ with equal total weight ▷ label-free
if $\mathcal{K} \neq \emptyset$ **then** ▷ few-label shift (Section 3)
  $\hat{b} \leftarrow \arg\max_b \sum_{i \in \mathcal{K}} \log \mathrm{Bern}\big(W_i;\, \sigma(r_i + b)\big) - \frac{b^2}{2}$, $r_i = \mathrm{logit}\,\hat{\rho}(z_i)$

Algorithm 3 Inference

Require: input x, f, T, $\hat{\pi}$, $\hat{\rho}$, λ, mean avoidable cost $\bar{c}$ of $f(T(x))$
Ensure: label $\hat{y}$, action g, score $\hat{B}$
$p_0 \leftarrow f(x)$; $(x', m) \leftarrow T(x)$; $z \leftarrow z(x, p_0, m)$
$\hat{B} \leftarrow \hat{\pi}(z)\,(2\hat{\rho}(z) - 1)$; $g \leftarrow \mathbf{1}[\hat{B} > \lambda\bar{c}]$
if $g = 1$ **then** $\hat{y} \leftarrow \mathbf{1}[f(x') > 1/2]$ ▷ second pass
else $\hat{y} \leftarrow \mathbf{1}[p_0 > 1/2]$
return $\hat{y}, g, \hat{B}$

5 Experimental Setup

Data. Table 1 lists the sets. The classifier is trained on Arabic-script TSAC (Medhaffar et al., 2017); the human Latin-script TSAC training comments, never seen by f, form the policy pool. Test sets cover the same corpus (S_3), another Tunisian corpus and platform (S_4, TUNIZI; Fourati et al., 2021), Algerian (S_5, NArabizi; Seddah et al., 2020; Touileb and Barnes, 2021) and Moroccan Arabizi (S_6, MYC; Jbel et al., 2025). Mixed-script sentences (minority alphabet $> 20\%$ of letters) are excluded. Each row is its own resampling unit except the S_5/S_5^{AR} pairs, which share a source identifier.

Set	Variety, script	N	Pos.%	Role
TSAC-AR	Tun., Arabic	5,564	45.0	f (90/10)
TSAC-LAT tr.	Tun., Latin	6,204	57.3	policy
S_3 TSAC-LAT	Tun., Latin	1,425	55.2	test
S_4 TUNIZI	Tun., Latin	2,875	48.8	test
S_5 NArabizi	Alg., Latin	729	52.0	test
S_5^{AR} NArabizi gold	Alg., Arabic	729	52.0	RQ4
S_6 MYC-LAT	Mor., Latin	4,191	66.4	test

Table 1: Data after de-duplication and near-copy removal (character 3-gram cosine ≥ 0.9 against every training or policy sentence). S_5^{AR} is the human Arabic-script transliteration of S_5 (Touileb and Barnes, 2021).

Corpora were cloned at fixed commits and are not redistributed (Appendix B).

Classifiers. f_θ is a logistic regression on mean-pooled, frozen representations of four encoders: XLM-R base (Conneau et al., 2020), mE5 base (Wang et al., 2024), LaBSE (Feng et al., 2022) and MARBERTv2 (Abdul-Mageed et al., 2021), pinned to fixed revisions. The regularization constant is chosen and a temperature is fitted (Guo et al., 2017) on the 10% Arabic-script development split. No encoder is fine-tuned, and no Latin-script sentence is used to train f.

Systems. B0 keeps x; B1 always normalizes with T_{cons}; B2 routes at random at SelectNorm's selection rate (20 draws); B3 thresholds the confidence of p_0; B4 estimates $B(z)$ directly as $P(\mathrm{help} \mid z) - P(\mathrm{harm} \mid z)$, a post-hoc deferral rule (Jitkrittum et al., 2023); B4b predicts each view's error (Kim et al., 2020); B6 is the oracle; R averages both views' probabilities (always two passes).

SelectNorm-src uses Equation (3) with source estimates; SelectNorm-adapt re-estimates $\hat{\pi}$ on unlabeled target text with two-fold cross-fitting by source sentence. All thresholds, including SelectNorm's, are tuned on the policy development split for each λ (the protocol's primary variant); SelectNorm with the untuned Bayes threshold $\lambda\bar{c}$ is reported as a variant. Nothing is tuned on a test set.

Classifier regimes. In the primary regime f sees only Arabic script. As an exploratory second regime, logged as a deviation before it was run, f is also trained on a synthetic Latin rendering $G(x)$ of each training sentence, produced by inverting the TArC lexicon and the digit rules (+SYN); this makes both views in-distribution to some degree and tests whether routing matters more when neither view dominates.

Few-label setting. This setting uses target labels and is reported separately. Each test set is split into two folds by source sentence. To predict fold f, a budget of $k \in \{8, 16, 32, 64, 128\}$ labels is drawn from the *other* fold only, either at random or among its disagreements (D is label-free); the labels re-estimate the shift b of $\hat{\rho}$ (SelectNorm), re-fit B4's heads, re-tune B3, or choose B0 or B1 for the whole fold (set-level choice). No sentence is scored by a system that saw its label; an audit that replays every targeted and random draw, checked against the indices logged by a rerun, found no sentence whose own label informed its prediction (Appendix B). On S_5 at k=128 a fold has only 97–116 disagreements, so the targeted budget is smaller there. Each k uses 5 draws $\times$ 5 seeds; all few-label systems use the $\lambda = 0$ threshold, and the zero-shot reference is SelectNorm-adapt with the same threshold.

Mechanism analysis. To locate failures we measure four quantities separately for each encoder and test set (Section 6.3): (M1) how well $\hat{\pi}$ ranks disagreements (AUC; source-trained and label-free target cross-fitted); (M2) how well $\hat{\rho}$ ranks the winning view among disagreements (AUC), its calibration gap, and a diagnostic ceiling obtained by cross-validating $\hat{\rho}$ on *target* labels (5 folds; not deployable); (M3) the macro-F1 gain over the best single view of the λ=0 rule $\hat{\rho} > 1/2$ under both estimates; (M4) the oracle headroom, compared with a *reference*: the headroom between two classifiers trained on bootstrap resamples of the same data and applied to the same original input. The reference shows how large "choose whichever is right" gains can be from training-sample variability alone; it is a diagnostic, not a decomposition of the headroom, and we report its sensitivity (Table 11). Test labels serve only as evaluation targets. Intervals for M1–M3 are 95% percentile intervals from 1,000 bootstrap resamples of test sentences; each resample reweights all five seeds' predictions with the same counts and averages over seeds, so repeated measurements are not treated as independent; estimators are not refitted, so the intervals cover evaluation sampling only.

Metrics and statistics. Macro-F1 is primary; we also report accuracy, selection rate, harmful transformations (a correct original prediction made wrong) per 100 inputs, the share of oracle opportunities captured, and inference cost relative to one pass of f on x, charging each system for what it runs (estimated from a token-length model fitted to measured latencies, and checked by end-to-end timing; Appendix B) (Figure 2). Results average five seeds. f is trained once per encoder and regime; seed s fixes the policy split (hence routers and thresholds), B2's draws and the few-label draws. Fits are deterministic, so seed variation reflects the policy split only, not retraining of f. Differences are tested with a paired bootstrap over source sentences (Koehn, 2004; Dror et al., 2018) (10,000 resamples, seed-averaged statistic), with Holm correction over all main comparisons of an encoder. We call a difference practically relevant only if it is at least one macro-F1 point and significant after correction (the protocol's wording requires only the unadjusted interval to exclude zero; the two coincide in the primary regime, and for +SYN we report both). A non-significant difference is not read as equivalence; where we say two systems perform alike, we give the paired difference and its interval (Table 7).

Status of analyses. Pre-specified in the protocol (Appendix B): main comparisons and statistics, the λ grid, the embedding, boosting and T_{lex} ablations, the few-label setting and RQ4; other ablations use the pre-registered code but are not named in it. Exploratory, logged after the first mE5 results: T_{cov}, +SYN and M1–M4. Added in revision: intervals, sensitivity and equivalence readings, end-to-end timing, the normalizer audit and the input-dependent cost threshold.

	mE5				XLM-R			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	63.4$^{\triangledown}$	61.5$^{\triangledown}$	63.6$^{\triangledown}$	**54.1**	54.5$^{\triangledown}$	48.8$^{\triangledown}$	48.6$^{\triangledown}$	63.5$^{\triangledown}$
B1 always normalize	**82.0**	**67.2**	69.4	52.9	72.8	**64.5**	63.1	72.3$^{\blacktriangle}$
B2 random, matched rate	80.0	66.5	68.5	53.2	70.2	63.3	61.9	68.9
B3 confidence threshold	82.0	67.2	69.4	52.9	72.9	64.4	63.0	**72.3**$^{\blacktriangle}$
B4 direct gain	81.4	67.0	69.9	53.6	73.5	63.9	63.3	68.7
B4b per-view loss	81.6	67.1	69.9	53.7	**73.9**	63.9	**63.4**	68.2
SelectNorm-src	81.3	67.0	**70.0**	53.7	73.8	63.8	63.4$^{\blacktriangle}$	68.0$^{\triangledown}$
SelectNorm-adapt	81.3	66.9	69.8	53.7	73.8	63.8	63.1	68.5
Bayes threshold $\lambda\bar{c}$	81.4	66.9	69.6	53.5	73.9	64.0	63.4	69.0
R two-view ensemble†	78.9	66.5	69.7	54.4	67.3	57.2	54.7	69.7
B6 oracle‡	90.0	81.3	81.8	63.8	83.1	75.2	70.4	82.1
	LaBSE				**MARBERTv2**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	69.6$^{\triangledown}$	61.1$^{\triangledown}$	65.6$^{\triangledown}$	**63.4**$^{\blacktriangle}$	64.3$^{\triangledown}$	62.4$^{\triangledown}$	60.0	52.5$^{\triangledown}$
B1 always normalize	83.6	**69.1**	70.9	55.5$^{\triangledown}$	82.1	67.7	62.3	58.3
B2 random, matched rate	81.0	67.3	69.5	60.4	79.9	67.3	61.9	56.7
B3 confidence threshold	83.7	69.1	70.9	55.9$^{\triangledown}$	**82.3**	68.1	62.3	**58.3**
B4 direct gain	83.4	68.6	71.6	60.6	82.1	68.2	**63.0**	57.6
B4b per-view loss	83.3	68.9	**71.6**	60.5	82.1	68.1	62.8	57.6
SelectNorm-src	83.4	68.7	71.1	60.4$^{\triangledown}$	82.2	68.3	62.9	57.7
SelectNorm-adapt	83.4	68.8	71.5	60.9	82.1	**68.3**	62.9	57.7
Bayes threshold $\lambda\bar{c}$	**83.7**	68.8	71.3	60.8	82.2	68.1	62.5	57.9
R two-view ensemble†	81.7	68.4	72.1	60.5	82.3	69.7	62.4	58.4
B6 oracle‡	91.5	82.2	85.7	75.2	93.6	84.2	77.6	70.4

Table 2: Macro-F1 ($\lambda = 0$, mean of five seeds), classifier trained on Arabic script only; two encoders per panel. Bold: best deployable system per column. Shaded: SelectNorm-adapt, the protocol's primary system. $^{\triangledown}$/$^{\blacktriangle}$: the baseline is significantly worse/better than SelectNorm-adapt (paired bootstrap, Holm-corrected $p < 0.05$ over the protocol's 20 comparisons per encoder; tested only for B0, B1, B3, B4 and SelectNorm-src). R always runs both passes; B6 uses test labels.

6 Results

6.1 Selective normalization against strong baselines (RQ1, RQ2)

Table 2 gives the main comparison. Normalization itself matters: always normalizing (B1) improves over the original input on the three Tunisian and Algerian sets for every encoder, by 2.3 to 18.6 macro-F1 points. On Moroccan MYC the sign depends on the encoder: B1 gains 8.8 points with XLM-R and loses 7.9 with LaBSE.

Learned routing does not add a consistent improvement on top of this. Against the direct-gain router B4 no difference is significant, and every 95% interval lies inside $[-0.63, +0.58]$, i.e. within the protocol's 1-point relevance margin in all 16 cells (an equivalence reading added post hoc; Table 7). SelectNorm-adapt and -src differ by -0.25 to $+0.54$ points (three small differences are significant), and all learned routers are descriptively within about one point of each other. Against B1 and B3, SelectNorm-adapt is relevantly better in one cell (LaBSE on MYC, $+5.5$ and $+5.0$), where normalizing is harmful and confidence thresholding fails to stop it, and relevantly worse in one (XLM-R on MYC, -3.8), where normalizing is right and the router keeps the original view for about 40% of inputs; in the LaBSE MYC cell it is also relevantly worse than B0, the better view there (-2.5). Elsewhere, non-significance is not equivalence: only 4 (B1) and 6 (B3) of the 16 intervals lie within ± 1 point, and the other non-significant ones extend from -1.3 to $+2.4$, so a small gain or loss cannot be excluded. Under the decision rule fixed for this study, SelectNorm therefore shows no consistent, statistically supported gain over competitive baselines; the rest of this section analyses why.

6.2 Accuracy and cost (RQ3)

Each single-view system runs one encoder pass: B0 costs 1.00 and B1 0.97–1.04 passes on x (the normalizer adds about 0.4%). A router that decides from p_0 has already paid for the first pass: at $\lambda = 0$, SelectNorm-adapt costs 1.69–1.88 passes for mE5 and 1.43–1.88 across encoders, against 1.89–2.05 for the two-view ensemble. Raising λ

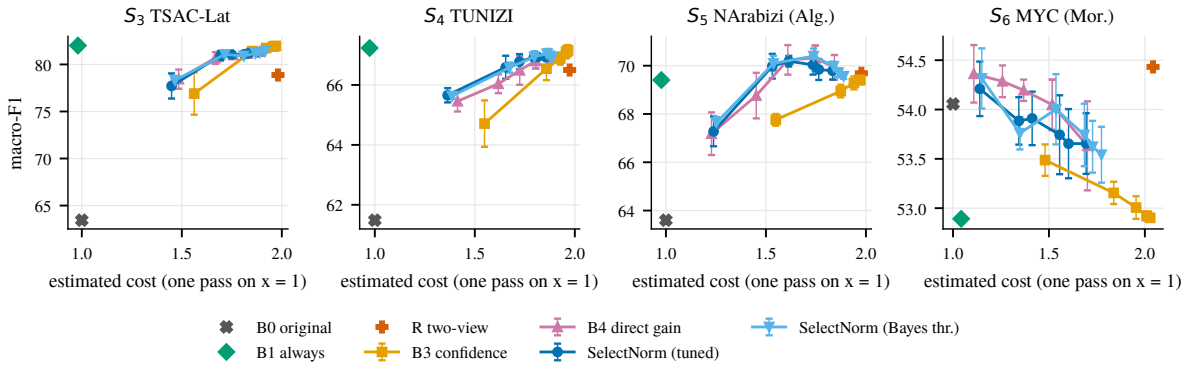


Figure 2: Macro-F1 against estimated inference cost (mE5, batch size 1, CPU; one pass of f on x = 1; per-sentence encoder costs from a token-length model fitted to measured latencies, see Appendix B), for $\lambda \in \{0, .01, .02, .05, .1, .2\}$; error bars: s.d. over seeds. Each system is charged for what it runs: B0 one pass on x; B1 the normalizer and one pass on $T(x)$; learned routers the first pass and the normalizer (sunk) plus the second pass when selected; B3 the first pass plus normalizer and second pass when selected; R both passes.

lowers the cost (to 1.14–1.45 for mE5 at $\lambda = 0.2$), but macro-F1 falls towards that of B0 (Figure 2). Per-input routing of this kind is thus never cheaper than the better single view and, never being relevantly more accurate (at most $+0.6$ points, not significant), offers no accuracy–cost advantage over it; it saves compute only relative to running both views. Thresholding with each input's own avoidable cost instead of the mean only moves points along the same frontier. These costs are token-model estimates; running the deployed pipeline end to end, with live routing decisions that reproduce the archived ones exactly, gives the same picture (B1 0.91–1.05, SelectNorm-adapt 1.31–1.97, R 1.91–2.07 passes; Appendix B). Descriptively, B4 and B3 trace similar curves, and the untuned Bayes threshold a curve close to that of tuned thresholds.

6.3 Where per-input routing fails (RQ5)

The factorization lets us measure each sub-problem separately (Table 3).

(M1) Disagreement is moderately predictable. Between 21% and 39% of inputs change label under normalization, and the label-free estimate ranks them with AUC 0.62–0.74; every 95% interval lies between 0.57 and 0.77 (Table 10). Re-estimating $\hat{\pi}$ on unlabeled target text raises this AUC most on MYC (for mE5 from 0.67 with source training to 0.74), but the Bayes rule at $\lambda = 0$ depends on $\hat{\rho}$ alone (Section 3); accordingly SelectNorm-src and -adapt differ by at most about half a point at $\lambda = 0$.

(M2) Direction is weakly predictable from z, even in domain. Which view is right when they disagree is ranked by source-trained $\hat{\rho}$ with AUC 0.54–0.71 in 14 of 16 cells (interval upper bounds at most 0.76); for XLM-R on Algerian text the interval includes 0.5, and on Moroccan text it lies below it ($[0.41, 0.48]$). Fitting $\hat{\rho}$ on target labels by cross-validation, an upper reference that no deployable system has, reaches 0.53–0.68 (upper bounds at most 0.73), and its interval includes 0.5 in five cells. With linear estimators, the decision-point features thus carry little information about direction in any domain we tested; this does not exclude that other features or estimators would predict it better. What does change across domains is the *level* of ρ: the win rate ranges from 33% to 77% across encoder–set pairs, and source $\hat{\rho}$ is miscalibrated out of domain, with gaps whose intervals exclude zero in 11 of 12 out-of-domain cells (up to $+20$ points).

(M3) The expected benefit follows. The rule $\hat{\rho} > 1/2$ gains between -3.3 and $+1.0$ points over the best single view with source estimates; only one gain has an interval above zero (XLM-R, S_3, $+1.0$), and three losses lie below zero. With the target-CV ceiling the change is -0.7 to $+2.8$, with interval upper bounds of at most $+3.9$. Most of the gap between routing and the oracle is therefore not a transfer problem: with these features and a linear estimator, it remains when the router is fitted on the target.

(M4) A reference for the oracle headroom. The oracle gains 7–17 points over the best view, which looks like large headroom. Two classifiers that differ only in a bootstrap resample of their training data, applied to the same input, show 3–15 points

		(M1) disagreement		(M2) direction				(M3) benefit		(M4) headroom	
Enc.	Set	$P(D)$	AUC_π	$P(W\|D)$	AUC_ρ	$\text{AUC}_\rho^{\text{tgt}}$	gap	Δ_{src}	Δ_{tgt}	oracle	vs. ref.
mE5	S_3	34	0.68	77	0.66	0.64	+3	-0.6*	+0.0	8.0	-0.3
	S_4	33	0.64	57	0.56	0.58	+19*	-0.3	-0.1	14.1	+4.6
	S_5	29	0.62	58	0.63	0.57°	+15*	+0.2	+0.7	12.4	+3.2
	S_6	21	0.74	47	0.54	0.56	+20*	-0.5	+0.2	9.7	+4.6
XLM-R	S_3	36	0.64	72	0.71	0.68	+3	+1.0*	+0.8	10.3	+5.9
	S_4	31	0.66	64	0.57	0.57	+11*	-0.5	+0.3	10.7	+7.6
	S_5	23	0.69	69	0.45°	0.60°	+11*	+0.3	-0.7	7.3	+2.1
	S_6	28	0.72	69	0.44	0.66	-8*	-3.3*	+1.1*	9.8	+4.5
LaBSE	S_3	30	0.69	73	0.61	0.53°	-0	+0.1	-0.1	7.9	+1.3
	S_4	34	0.65	62	0.57	0.59	+6*	-0.4	-0.4	13.0	+5.7
	S_5	34	0.65	58	0.59	0.55°	+3	+0.4	+0.2	14.9	+7.5
	S_6	31	0.68	33	0.57	0.64	+16*	-2.6*	+0.8*	11.8	+5.3
MARBERTv2	S_3	39	0.69	70	0.63	0.60	+1	+0.1	-0.3	11.5	+0.8
	S_4	37	0.71	60	0.61	0.64	+13*	+0.4	+2.8*	16.5	+2.0
	S_5	31	0.73	57	0.63	0.54°	+13*	+0.2	+1.1	15.3	+3.6
	S_6	30	0.71	59	0.55	0.63	+4*	-0.3	+1.1*	12.1	-0.7

Table 3: The four mechanisms of Section 3 (T_{cons}, Arabic-script classifier, mean of five seeds). (M1) disagreement rate (%) and AUC of label-free, target-cross-fitted $\hat{\pi}$. (M2) win rate of $T(x)$ on disagreements (%), AUC of source-trained $\hat{\rho}$, AUC of a diagnostic $\hat{\rho}$ cross-validated on target labels (not deployable), and calibration gap of source $\hat{\rho}$ (points). (M3) macro-F1 gain over the best single view of the rule $\hat{\rho} > 1/2$ with source and with target-CV $\hat{\rho}$. (M4) oracle headroom over the best view, and the same minus the headroom between two bootstrap-resampled classifiers on the same input (a diagnostic reference, not a decomposition; see Table 11). °: the 95% interval of the AUC contains 0.5; *: the interval of the gap or gain excludes 0 (1,000 seed-averaged bootstrap resamples; all intervals in Table 10). Test labels are used for analysis only.

of the same kind of headroom without any normalization. This reference is a diagnostic, not a decomposition: it depends strongly on how the two classifiers are made (2–6 points with 80% subsamples, 5–22 points with weaker regularization; Table 11), so it cannot say how much of the views' headroom is recoverable. It does show that oracle headroom of this size arises from training-sample variability alone, and that oracle headroom is a weak indicator of what a router can gain.

Linguistic conditions. On the Tunisian and Algerian sets, always normalizing helps less on inputs with many French or English tokens or low lexicon coverage, for every encoder but one case (XLM-R, TUNIZI, French/English) (Table 4); negation and length show no consistent pattern. On MYC the net effect of every condition has a different sign for different encoders, so the main determinant of whether normalization pays there is the encoder–dialect pair, not a property of the input that z could observe.

6.4 Few target labels (RQ6)

With k labelled target sentences, re-estimating the scalar shift of $\hat{\rho}$ changes macro-F1 relative to zero-shot SelectNorm-adapt with the same threshold by -1.4 to $+1.5$ points across encoders, regimes, sets and k (mean -0.01; Figure 3). Spending the labels on disagreements, which Proposition 2 shows to be $1/P(D{=}1)$ times more informative about the shift, makes little difference: averaged over encoders and sets, SelectNorm with targeted labels is between -0.1 and $+0.2$ points from SelectNorm with random labels at every k (descriptive; not tested). In the Arabic-script regime, SelectNorm is clearly better than a set-level choice between B0 and B1 when $k \leq 16$ (by 2.6–2.9 points on average), and the advantage shrinks to 0.3 points at $k = 128$, consistent with M2: most of the usable signal is set-level.

6.5 Normalizer, consistency and classifier regime

Safeguards. The safeguards of T_{cons} matter more than routing: always normalizing with T_{lex} or T_{rule} instead loses 4–11 points on every set for mE5, and routing over those normalizers recovers only part of

Condition (share of S_3/S_4/S_5/S_6)	S_3	S_4	S_5	S_6
French/English $\geq$30% (9/12/12/22)	+4.3 [+0,+11]	+4.3 [-2,+13]	+1.6 [-3,+5]	-0.1 [-9,+7]
not French/English $\geq$30% (91/88/88/78)	+17.0 [+15,+20]	+7.6 [+5,+9]	+6.5 [+5,+10]	+1.3 [-11,+12]
negation rule applied (8/12/26/6)	+14.0 [+6,+18]	+5.9 [-4,+20]	+4.0 [-4,+9]	+2.1 [-9,+15]
not negation rule applied (92/88/74/94)	+16.1 [+14,+18]	+7.4 [+6,+11]	+6.5 [+3,+11]	+1.0 [-11,+10]
lexicon coverage <50% (51/57/81/85)	+12.3 [+11,+14]	+4.5 [+2,+9]	+3.7 [+1,+6]	+0.8 [-11,+11]
not lexicon coverage <50% (49/43/19/15)	+19.7 [+17,+23]	+10.8 [+9,+14]	+15.1 [+8,+22]	+2.4 [-11,+13]
$\leq$3 tokens (42/28/2/27)	+14.2 [+10,+16]	+8.4 [+0,+15]	+13.2 [-18,+35]	+0.7 [-10,+10]
not $\leq$3 tokens (58/72/98/73)	+17.2 [+13,+20]	+6.8 [+3,+10]	+5.7 [+4,+10]	+1.2 [-11,+11]

Table 4: Net effect of always normalizing (help minus harm per 100 inputs) by input condition, mean over the four encoders, range in brackets. Conditions were fixed before inspection.

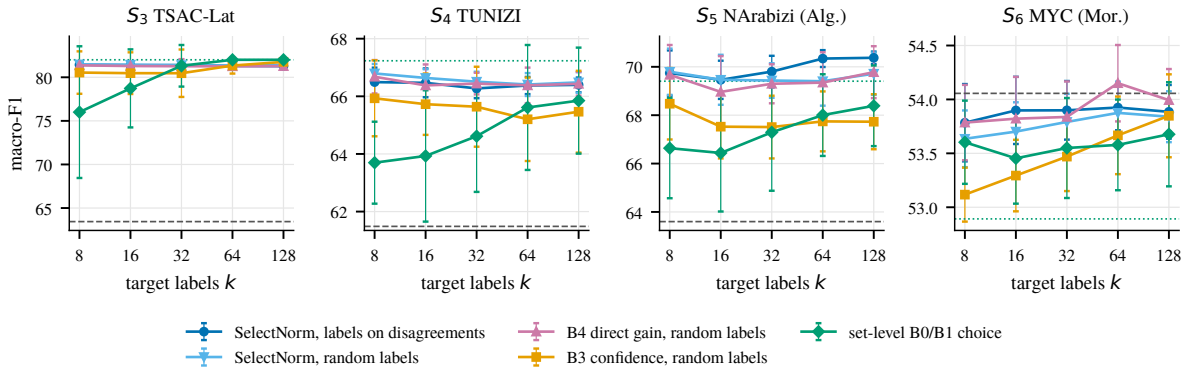


Figure 3: Few-label adaptation (mE5): macro-F1 against the number k of labelled target sentences (5 draws $\times$ 5 seeds; error bars: s.d.). Dashed: B0; dotted: B1. "Set-level choice" picks B0 or B1 by accuracy on the k labels.

the loss (Table 5). Keeping out-of-lexicon tokens in Latin script (T_{cov}) reduces disagreements by about a third and makes routing nearly irrelevant; it is worse than T_{cons} on the Tunisian sets and slightly better on S_5 and S_6.

Consistency with human transliteration (RQ4). On NArabizi, B1 agrees with the prediction that f makes on the human Arabic-script version of the same sentence for 73–77% of inputs, against 61–67% for B0 and 73–76% for SelectNorm-adapt, so routing does not increase agreement further; f itself reaches 65–77 macro-F1 on the human transliterations (per-encoder values in the released `consistency.csv`).

Classifier trained on both scripts. When f also sees synthetic Latin renderings (+SYN, Table 8), the original view improves and the two views become more complementary: the oracle headroom stays similar, the win rate on disagreements moves towards 50%, and SelectNorm-adapt exceeds the better single view in several cells. Against that view it is relevantly better once with Holm correction (MARBERTv2, S_4, $+1.1$) and twice more under the protocol's unadjusted criterion (mE5, S_3 $+2.1$, S_4 $+1.4$), and relevantly worse in two MYC cells under both (LaBSE -2.4, MARBERTv2 -1.8). Confidence thresholding and the ensemble share the gains; SelectNorm is never significantly better than B4 (worse once, -0.4).

7 Discussion

What the factorization shows. A per-input router can fail because it cannot tell *whether* the views disagree, *which* wins when they do, how the latter shifts across domains, or because the apparent headroom is not real; measured separately, these have different weights here. Disagreement is moderately predictable (M1). Transfer matters in specific cells: for XLM-R on Algerian and Moroccan text, source $\hat{\rho}$ ranks direction at or below chance, and fitting it on the target turns a loss of up to 3.3 points into a small gain. But even fitted on the target, $\hat{\rho}$ gains at most 2.8 points over the best view in any cell (M2, M3). In our measurements the main limit is that the decision-point features say little about which view is right, even in domain, and a sizeable part of the oracle headroom is of the size that two classifiers differing only in their training sample also show (M4; a diagnostic reference, not a decomposition). The richer variants we tried did

not change this (16 embedding components: -1.2 to $+1.1$ points, no consistent sign; gradient boosting, mE5 only: -0.6 to $+0.7$). These are limits of the features and estimators we evaluated; features computed after the second pass, fine-tuned representations or other estimators might do better, and our results do not show that the conditional advantage is intrinsically unpredictable. The same measurements can be made for other two-view choices with a frozen model, such as test-time augmentation or translate-test; we have not done so. Where oracle headroom is reported, a bootstrap-classifier reference helps to judge how much of it is classifier variability.

Where the signal is. The win rate on disagreements varies far more between encoder–dialect pairs (33–77%) than our features resolve within a set, and on MYC every linguistic condition changes sign with the encoder. The useful decision is thus mostly set-level, which few labels resolve: with 64–128 target labels a set-level choice is on average within 0.3–0.7 points of SelectNorm with the same labels (descriptive). Targeted labels estimate the shift more efficiently (Proposition 2) but cannot fix a weak ranking.

Practical recommendations. For Arabizi with a frozen encoder, a linear head trained on Arabic script, and binary sentiment, our results suggest: (i) invest in normalizer safeguards first, since they matter more than routing (Table 5); (ii) choose between normalizing and not at the level of a corpus or dialect, with on the order of a hundred labels; (iii) do not expect per-input routing that reads the first pass to save compute, except relative to running both views; (iv) when the classifier is trained on both scripts, learned routers, confidence thresholding or an ensemble sometimes gain a little over the better view, but can also lose on MYC.

Relation to prior findings. That transliteration helps unseen scripts on average is known (Muller et al., 2021; Moosa et al., 2023); our results agree for Tunisian and Algerian Arabizi and add that the sign flips with the encoder on Moroccan text. Loss predictors for test-time transformations (Kim et al., 2020) and post-hoc deferral (Jitkrittum et al., 2023) are the closest methods; SelectNorm is within one point of the direct-gain router in every cell, and the factorization adds a diagnosis, not an accuracy gain.

Limitations

Scope of the task. All experiments use binary sentiment, where the factorization is exact with ρ alone; the K-class form needs two conditional win rates. **Classifiers and scope.** Encoders are frozen and heads are linear; all conclusions are restricted to this setting. A fine-tuned encoder could change how often normalization helps and how predictable the winning view is; we did not fine-tune (CPU only), and the exploratory +SYN regime is only a partial probe. **Normalizer.** T is built from Tunisian resources (TArC lexicon, Tunisian digit conventions such as 8=ه); its intrinsic accuracy is lower on Algerian NArabizi, and no Moroccan transliteration gold was available, so its validity on MYC is unmeasured. Safeguards were tuned on the TArC held-out split, and no independent transliteration test set was available, so intrinsic accuracy is not independently validated. **Data.** Cross-corpus differences mix dialect with platform, topic, period and label distribution; we do not attribute them to dialect alone. TUNIZI, NArabizi and MYC state no licence and MARBERTv2 has none on its model card; we use them for research only and do not redistribute them. **Development.** The pipeline was first exercised with a character n-gram classifier whose test results we inspected; this motivated the few-label setting. Prespecified and exploratory analyses are distinguished in Section 5. **Analysis.** Mechanism intervals cover resampling of test sentences, not refitting of the estimators; the target-label ceiling uses linear $\hat{\rho}$ on the same features, so a richer model could find more signal. Features computed after $f(T(x))$ might predict direction better but need the second pass on every input; we did not study them. Equivalence statements use 95% intervals against the protocol's 1-point margin, are post hoc and are not corrected for multiplicity. **Costs.** Per-sentence costs are estimated from a token-length model and checked by end-to-end timing of the deployed pipeline on 200 sentences per set, on one 4-core CPU without a GPU; relative costs on accelerators differ, especially for short inputs.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of ACL-IJCNLP*, pages 7088–7105.

Bashar Alhafni, Sarah Al-Towaity, Ziyad Fawzy, Fatema

Nassar, Fadhl Eryani, Houda Bouamor, and Nizar Habash. 2024. Exploiting dialect identification in automatic dialectal text normalization. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 42–54.

Mikel Artetxe, Vedanuj Goswami, Shruti Bhosale, Angela Fan, and Luke Zettlemoyer. 2023. Revisiting machine translation for cross-lingual classification. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6489–6499.

Niyati Bafna, Emily Chang, Nathaniel Romney Robinson, David R. Mortensen, Kenton Murray, David Yarowsky, and Hale Sirin. 2025. DialUp! modeling the language continuum by adapting models to dialects and dialects to models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 20188–20233.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of ACL*, pages 8440–8451.

Kareem Darwish. 2014. Arabizi detection and conversion to Arabic. In *Proceedings of the EMNLP 2014 Workshop on Arabic Natural Language Processing (ANLP)*, pages 217–224.

Antonios Dimakis, John Pavlopoulos, and Antonios Anastasopoulos. 2025. Dialect normalization using large language models and morphological rules. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23696–23714.

Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. The hitchhiker's guide to testing statistical significance in natural language processing. In *Proceedings of ACL*, pages 1383–1392.

Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2022. Language-agnostic BERT sentence embedding. In *Proceedings of ACL*, pages 878–891.

Chayma Fourati, Hatem Haddad, Abir Messaoudi, Moez BenHajhmida, Aymen Ben Elhaj Mabrouk, and Malek Naski. 2021. Introducing a large Tunisian Arabizi dialectal dataset for sentiment analysis. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 226–230.

Saurabh Garg, Sivaraman Balakrishnan, Zachary Chase Lipton, Behnam Neyshabur, and Hanie Sedghi. 2022. Leveraging unlabeled data to predict out-of-distribution performance. In *International Conference on Learning Representations*.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, pages 4878–4887.

Yonatan Geifman and Ran El-Yaniv. 2019. SelectiveNet: A deep neural network with an integrated reject option. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *PMLR*, pages 2151–2159.

Elisa Gugliotta and Marco Dinarelli. 2022. TArC: Tunisian Arabish corpus, first complete release. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 1125–1136.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of ICML*, pages 1321–1330.

Neha Gupta, Harikrishna Narasimhan, Ankit Singh Rawat, Wittawat Jitkrittum, Aditya Krishna Menon, and Sanjiv Kumar. 2024. Language model cascades: Token-level uncertainty and beyond. In *International Conference on Learning Representations*.

Mouad Jbel, Mourad Jabrane, Imad Hafidi, and Abdulmutallib Metrane. 2025. Sentiment analysis dataset in Moroccan dialect: Bridging the gap between Arabic and Latin scripted dialect. *Language Resources and Evaluation*, 59(2):1401–1430. Published online 2024; preprint arXiv:2303.15987.

Wittawat Jitkrittum, Neha Gupta, Aditya K. Menon, Harikrishna Narasimhan, Ankit Singh Rawat, and Sanjiv Kumar. 2023. When does confidence-based cascade deferral suffice? In *Advances in Neural Information Processing Systems*.

Deokhyung Kang, Gary Geunbae Lee, and Hyounghun Kim. 2026. Learning when to translate for multilingual reasoning. arXiv:2606.02465.

Ildoo Kim, Younghoon Kim, and Sungwoong Kim. 2020. Learning loss for test-time augmentation. In *Advances in Neural Information Processing Systems*, volume 33, pages 4163–4174.

Philipp Koehn. 2004. Statistical significance tests for machine translation evaluation. In *Proceedings of EMNLP*, pages 388–395.

Salima Medhaffar, Fethi Bougares, Yannick Estève, and Lamia Hadrich-Belguith. 2017. Sentiment analysis of Tunisian dialects: Linguistic resources and experiments. In *Proceedings of the Third Arabic Natural Language Processing Workshop*, pages 55–61.

Ibraheem Muhammad Moosa, Mahmud Elahi Akhter, and Ashfia Binte Habib. 2023. Does transliteration help multilingual language modeling? In *Findings of the Association for Computational Linguistics: EACL 2023*, pages 670–685.

Hussein Mozannar and David Sontag. 2020. Consistent estimators for learning to defer to an expert. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *PMLR*, pages 7076–7087.

Benjamin Muller, Antonios Anastasopoulos, Benoît Sagot, and Djamé Seddah. 2021. When being unseen from mBERT is just the beginning: Handling new languages with multilingual language models. In *Proceedings of NAACL-HLT*, pages 448–462.

Sima Noorani, Shayan Kiyani, George Pappas, and Hamed Hassani. 2026. Disagreement-guided weak-to-strong routing. arXiv:2610.22245.

Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. 2025. RouteLLM: Learning to route LLMs from preference data. In *International Conference on Learning Representations*.

Emmanouil Antonios Platanios, Avrim Blum, and Tom M. Mitchell. 2014. Estimating accuracy from unlabeled data. In *Proceedings of the Thirtieth Conference on Uncertainty in Artificial Intelligence*, pages 682–691.

Djamé Seddah, Farah Essaidi, Amal Fethi, Matthieu Futeral, Benjamin Muller, Pedro Javier Ortiz Suárez, Benoît Sagot, and Abhishek Srivastava. 2020. Building a user-generated content North-African Arabizi treebank: Tackling hell. In *Proceedings of ACL*, pages 1139–1150.

H. Sebastian Seung, Manfred Opper, and Haim Sompolinsky. 1992. Query by committee. In *Proceedings of the Fifth Annual Workshop on Computational Learning Theory*.

Ali Shazal, Aiza Usman, and Nizar Habash. 2020. A unified model for Arabizi detection and transliteration using sequence-to-sequence models. In *Proceedings of the Fifth Arabic Natural Language Processing Workshop*, pages 167–177.

Yassine Toughrai, Kamel Smaïli, and David Langlois. 2025. Modeling North African dialects from standard languages. In *Proceedings of The Third Arabic Natural Language Processing Conference*, pages 375–383.

Samia Touileb and Jeremy Barnes. 2021. The interplay between language similarity and script on a novel multi-layer Algerian dialect corpus. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3700–3712.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.

A Proofs

Bayes rule (Equation (2)). For fixed z, the conditional risk of $g(z) = a$ is $\mathbb{E}[e_a \mid z] + \lambda\,\mathbb{E}[c_0 + c_T \mid z] + \lambda\bar{c}(z)\,a$, where the middle term does not depend on a. Choosing $a = 1$ is strictly better iff $\mathbb{E}[e_1 \mid z] + \lambda\bar{c}(z) < \mathbb{E}[e_0 \mid z]$, i.e. $B(z) > \lambda\bar{c}(z)$. Minimizing pointwise minimizes Equation (1). □

Variant	S_3	S_4	S_5	S_6
SelectNorm-adapt (full)	81.3	66.9	69.8	53.7
– target re-estimation of $\hat{\pi}$	81.3	67.0	70.0	53.7
Bayes threshold instead of tuned	81.4	66.9	69.6	53.5
$2\hat{\rho} - 1$ only (no $\hat{\pi}$)	81.3	66.9	69.8	53.6
$\hat{\pi}$ only, tuned (label-free ranking)	81.9	67.2	69.5	53.1
confidence features only	82.0	67.2	69.4	53.0
– confidence features	81.5	66.9	69.6	53.6
– script features	81.6	67.1	69.8	53.4
– normalizer reliability	82.0	67.0	69.4	53.2
+ embedding PCA	82.0	67.2	70.9	53.8
gradient boosting	82.0	67.1	69.6	53.1
T_{lex} (no safeguards)	80.7	64.6	65.2	53.0
always T_{lex}	77.8	61.8	63.2	47.9
T_{rule}	76.7	61.8	61.4	53.3
always T_{rule}	75.1	58.8	58.8	47.7
T_{cov} (unknown tokens kept Latin)	75.0	65.7	70.8	54.1
always T_{cov}	75.0	65.8	70.7	54.0

Table 5: Ablations (mE5, $\lambda = 0$, macro-F1, mean of five seeds). Rows with "always" are B1 with that normalizer.

	TArC development split (Tunisian)			NArabizi (Algerian, out of dialect)	
Strategy	Word acc.	Foreign kept (P/R)	Negation	Word acc.	Foreign kept (P/R)
T_{rule}	36.3	–/0	29.1	39.0	–/0
T_{lex}	69.5	–/0	54.4	47.6	–/0
T_{cons}	69.1	96/79	96.1	46.8	93/53

Table 6: Normalizer accuracy against human Arabic-script transliterations: *development* results, not an independent test. TArC: 20% of sentences, not used for the lexicon $\mathcal{L}$ but used to design the safeguards. NArabizi: whole corpus, nothing fitted to it, but scored during development. "Foreign kept": precision/recall of keeping annotated French/English tokens in Latin script; negation: accuracy on circumfix-negated verbs. Audit: no hand-listed word occurs only in the development split; on the 87% of tokens no rule touches, word accuracy is 65.8%; with the splits' roles swapped, negation is kept in 94.4% and foreign precision/recall is 95/77.

Factorization (Equation (3)). If $D = 0$ then $\hat{y}_0 = \hat{y}_1$ and $e_0 = e_1$, so $e_0 - e_1 = D(e_0 - e_1)$. If $D = 1$ and $y \in \{0, 1\}$, then $\hat{y}_0 = 1 - \hat{y}_1$, hence $e_0 = W$ and $e_1 = 1 - W$, and $e_0 - e_1 = 2W - 1$. Therefore $B(z) = \mathbb{E}[D(2W - 1) \mid z] = P(D{=}1 \mid z)\,\big(2P(W{=}1 \mid D{=}1, z) - 1\big)$. For K classes, $e_0 - e_1 = D\big(\mathbf{1}[\hat{y}_1{=}y] - \mathbf{1}[\hat{y}_0{=}y]\big)$, which gives $\pi(\rho_1 - \rho_0)$. □

Proposition 1. For any z, if $\hat{g}(z) = g^\star(z)$ the conditional excess risk is zero. Otherwise it equals $|B(z) - \lambda\bar{c}(z)|$, and since $\hat{B} - \lambda\bar{c}$ and $B - \lambda\bar{c}$ have different signs (or $\hat{B} = \lambda\bar{c}$), $|B - \lambda\bar{c}| \le |\hat{B} - B|$. Taking expectations gives $R(\hat{g}) - R(g^\star) \le \mathbb{E}|\hat{B} - B|$. Writing $\hat{B} - B = \hat{\pi}(2\hat{\rho} - 1) - \pi(2\rho - 1) = (\hat{\pi} - \pi)(2\rho - 1) + 2\hat{\pi}(\hat{\rho} - \rho)$ and applying the triangle inequality concludes. □

Proposition 2. Write $r_i = \operatorname{logit}\hat{\rho}(z_i)$ and $q_i = \sigma(r_i + b)$. A labelled target example contributes the log-likelihood $D_i\big[W_i \log q_i + (1 - W_i)\log(1 - q_i)\big]$: when $D_i = 0$ the label is uninformative about b because $e_0 = e_1$ regardless of y_i, and the model of W is defined only on disagreements. The per-example Fisher information about b is $D_i\, q_i(1 - q_i)$. Under random sampling from Q, its expectation is $P_Q(D{=}1)\,\mathbb{E}_Q[q(1 - q) \mid D{=}1]$; under sampling from

Enc.	Set	vs. B1	vs. B3	vs. B4	vs. SN-src
mE5	S_3	-0.71 [-1.25, -0.21]	-0.67 [-1.21, -0.17]	-0.07 [-0.27, +0.13]†	-0.03 [-0.11, +0.06]†
	S_4	-0.31 [-0.80, +0.17]†	-0.23 [-0.72, +0.25]†	-0.03 [-0.25, +0.18]†	-0.04 [-0.16, +0.08]†
	S_5	+0.39 [-0.22, +1.01]	+0.39 [-0.22, +1.01]	-0.14 [-0.57, +0.28]†	-0.17 [-0.47, +0.11]†
	S_6	+0.76 [+0.23, +1.30]	+0.75 [+0.23, +1.29]	+0.02 [-0.22, +0.27]†	-0.04 [-0.18, +0.11]†
XLM-R	S_3	+0.99 [+0.01, +2.02]	+0.91 [-0.07, +1.91]	+0.28 [+0.04, +0.52]†	+0.07 [-0.02, +0.17]†
	S_4	-0.66 [-1.20, -0.11]	-0.60 [-1.14, -0.04]	-0.04 [-0.18, +0.10]†	+0.02 [-0.04, +0.09]†
	S_5	-0.01 [-0.42, +0.45]†	+0.09 [-0.34, +0.56]†	-0.18 [-0.41, +0.03]†	-0.25 [-0.45, -0.07]$^{*\,\dagger}$
	S_6	-3.76 [-4.58, -2.93]*	-3.77 [-4.58, -2.94]*	-0.23 [-0.42, -0.03]†	+0.53 [+0.33, +0.73]$^{*\,\dagger}$
LaBSE	S_3	-0.19 [-0.75, +0.35]†	-0.24 [-0.73, +0.23]†	+0.03 [-0.19, +0.24]†	+0.03 [-0.08, +0.14]†
	S_4	-0.37 [-1.01, +0.28]	-0.35 [-0.95, +0.25]†	+0.15 [-0.06, +0.34]†	+0.01 [-0.11, +0.14]†
	S_5	+0.62 [-1.20, +2.44]	+0.57 [-1.17, +2.32]	-0.07 [-0.60, +0.47]†	+0.34 [-0.01, +0.71]†
	S_6	+5.46 [+4.44, +6.50]*	+5.04 [+4.05, +6.04]*	+0.32 [+0.06, +0.58]†	+0.54 [+0.30, +0.79]$^{*\,\dagger}$
MARBERTv2	S_3	-0.01 [-0.65, +0.64]†	-0.16 [-0.77, +0.48]†	+0.03 [-0.18, +0.23]†	-0.09 [-0.20, +0.01]†
	S_4	+0.57 [+0.08, +1.06]	+0.16 [-0.37, +0.68]†	+0.06 [-0.10, +0.22]†	+0.00 [-0.07, +0.08]†
	S_5	+0.58 [-0.45, +1.64]	+0.58 [-0.45, +1.64]	-0.11 [-0.63, +0.40]†	-0.01 [-0.36, +0.35]†
	S_6	-0.53 [-1.15, +0.08]	-0.61 [-1.22, -0.01]	+0.12 [-0.19, +0.43]†	-0.01 [-0.15, +0.12]†

Table 7: Paired differences in macro-F1 (SelectNorm-adapt minus baseline, $\lambda = 0$, seed-averaged) with 95% paired-bootstrap intervals (10,000 resamples of test sentences). *: Holm-corrected $p < 0.05$ over the protocol's 20 comparisons per encoder. †: the interval lies inside $(-1, +1)$, the protocol's relevance margin; reading this as equivalence is post hoc and not corrected for multiplicity. Intervals without either mark are inconclusive about both difference and equivalence.

	mE5+syn				**XLM-R+syn**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	80.9$^{\triangledown}$	65.6$^{\triangledown}$	69.1	**61.5**	69.0$^{\triangledown}$	59.8$^{\triangledown}$	64.7	59.7
B1 always normalize	81.6	67.5	72.6	58.0$^{\triangledown}$	72.8	63.2	62.9	59.1
B2 random, matched rate	81.2	66.5	70.4	61.0	70.8	61.6	63.9	59.7
B3 confidence threshold	83.0	68.7	**73.2**	60.7	74.5	63.0	64.7	60.3
B4 direct gain	**83.8**	**68.9**	72.4	60.9	**74.5**	63.8	65.1	60.6
B4b per-view loss	83.8	68.9	72.5	60.8	74.3	63.8	65.0	60.5
SelectNorm-src	83.6	68.8	72.9	61.1	74.3	63.5	**65.5**	60.4$^{\triangledown}$
SelectNorm-adapt	83.7	68.9	73.0	61.1	74.3	63.5	65.4	60.6
Bayes threshold $\lambda\bar{c}$	83.7	68.8	73.2	61.1	74.5	**63.9**	65.0	**60.8**
R two-view ensemble†	83.7	68.9	74.7	60.5	75.8	64.0	65.6	60.7
B6 oracle‡	91.7	80.9	85.7	71.4	86.3	79.0	80.6	73.5
	LaBSE+syn				**MARBERTv2+syn**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	76.6$^{\triangledown}$	64.9$^{\triangledown}$	71.3	**62.4**$^{\blacktriangle}$	77.0$^{\triangledown}$	64.2$^{\triangledown}$	57.0	50.1$^{\triangledown}$
B1 always normalize	83.7	67.8	74.3	58.7$^{\triangledown}$	81.8	67.8$^{\triangledown}$	59.3	**60.7**$^{\blacktriangle}$
B2 random, matched rate	80.3	66.3	72.9	59.9	80.8	67.3	59.1	58.3
B3 confidence threshold	**84.3**	68.2	74.3	60.4	82.9	68.7	59.3	59.5$^{\blacktriangle}$
B4 direct gain	84.3	**68.5**	74.4	59.7	82.5	68.7	59.6	59.3$^{\blacktriangle}$
B4b per-view loss	84.1	68.5	**74.8**	59.6	82.3	68.7	59.6	58.7
SelectNorm-src	84.1	68.3	74.7	60.0	82.5	68.8	59.8	59.0
SelectNorm-adapt	84.2	68.2	74.6	60.0	82.6	68.9	**59.9**	58.9
Bayes threshold $\lambda\bar{c}$	84.2	68.2	74.8	60.1	**83.0**	**69.2**	59.8	58.6
R two-view ensemble†	83.5	68.8	76.6	61.3	83.6	69.0	59.3	57.3
B6 oracle‡	92.9	81.1	86.8	73.1	92.6	81.6	72.3	69.3

Table 8: As Table 2, with f also trained on synthetic Latin renderings (+SYN, exploratory regime).

$Q(\cdot \mid D{=}1)$, which is possible without labels because D is observable, it is $\mathbb{E}_Q[q(1-q) \mid D{=}1]$. Information is additive over independent draws, so the ratio is $1/P_Q(D{=}1)$. □

The proposition concerns the scalar shift only. It does not claim that a shifted $\hat{\rho}$ recovers the target-optimal rule; that depends on whether the shift is close to a constant in logit space, which the experiments test.

B Reproducibility

Encoders and software. Encoders are loaded at pinned revisions with SHA-256-verified weights: XLM-R base `e73636d` (MIT), mE5 base `d128750` (MIT), LaBSE `836121a` (Apache-2.0), MARBERTv2 `fe88db9` (no licence stated). Representations: mean of the last layer over non-padding tokens, at most 128 subwords, ℓ_2-normalized; mE5 inputs carry `query:`. Python 3, scikit-learn 1.9.1, NumPy 2.5.3; code, configuration, logs and the protocol are released.

Classifier f. ℓ_2-regularized logistic regression; $C \in \{0.01, 0.03, \ldots, 100\}$ selected by macro-F1 on TSAC-AR dev; temperature fitted on the same split by NLL. The +SYN regime adds the synthetic Latin rendering $G(x)$ of every training and dev sentence.

Router. $\hat{\pi}$, $\hat{\rho}$: logistic regression on standardized z ($C = 1$); 17 features: confidence, margin and entropy of p_0 (CONF); length, Latin and Arabic shares, digit-as-letter, emoji and upper-case rates (SCRIPT); lexicon, rule-fallback, ambiguity, protected, negation, laughter and changed-token rates and lexicon entropy of T (TREL). Policy split 80/20 of TSAC-LAT train

	S_3 (help 376, harm 114)				S_4 (help 538, harm 402)				S_5 (help 124, harm 89)				S_6 (help 407, harm 456)			
System	Sel. %	Harm	Capt. %	Cost	Sel. %	Harm	Capt. %	Cost	Sel. %	Harm	Capt. %	Cost	Sel. %	Harm	Capt. %	Cost
B1	100.0	8.0	100.0	0.98	100.0	14.0	100.0	0.97	100.0	12.2	100.0	0.98	100.0	10.9	100.0	1.04
B3	98.9	8.0	99.7	1.97	99.1	14.0	99.5	1.97	99.5	12.2	100.0	1.97	98.5	10.9	99.9	2.03
B4	90.2	7.6	96.1	1.89	88.8	13.1	93.8	1.87	86.4	10.9	95.5	1.84	66.8	7.9	76.9	1.70
SN-adapt	89.0	7.6	95.7	1.88	87.0	13.0	93.3	1.86	85.7	11.3	96.8	1.83	65.9	8.3	80.9	1.69
Oracle	26.4	0.0	100.0	1.26	18.7	0.0	100.0	1.18	17.0	0.0	100.0	1.17	9.7	0.0	100.0	1.10

Table 9: Routing behaviour (mE5, $\lambda = 0$): selection rate, harmful transformations per 100 inputs, share of oracle opportunities captured, and estimated cost relative to B0 (Appendix B). Help/harm: number of inputs where normalization corrects/introduces an error.

Enc.	Set	AUC_π	AUC_ρ	$\mathrm{AUC}_\rho^{\mathrm{tgt}}$	gap (pts)	Δ_{src}	Δ_{tgt}
mE5	S_3	[0.65, 0.71]	[0.60, 0.72]	[0.58, 0.69]	[-0.8, +6.8]	[-1.0, -0.2]	[-0.8, +0.7]
	S_4	[0.62, 0.66]	[0.52, 0.60]	[0.54, 0.61]	[+15.2, +21.9]	[-0.8, +0.1]	[-1.2, +1.0]
	S_5	[0.57, 0.66]	[0.55, 0.71]	[0.49, 0.64]	[+7.7, +21.1]	[-0.2, +0.5]	[-1.5, +2.9]
	S_6	[0.73, 0.76]	[0.50, 0.57]	[0.52, 0.60]	[+16.7, +23.5]	[-1.7, +0.6]	[-0.7, +0.9]
XLM-R	S_3	[0.61, 0.67]	[0.66, 0.76]	[0.63, 0.73]	[-0.8, +6.2]	[+0.1, +2.0]	[-0.3, +2.0]
	S_4	[0.64, 0.68]	[0.53, 0.60]	[0.53, 0.61]	[+8.0, +14.6]	[-0.9, +0.0]	[-0.2, +0.8]
	S_5	[0.65, 0.74]	[0.35, 0.53]	[0.50, 0.69]	[+4.5, +18.0]	[+0.0, +0.7]	[-2.1, +0.8]
	S_6	[0.71, 0.74]	[0.41, 0.48]	[0.63, 0.70]	[-10.9, -4.7]	[-4.0, -2.5]	[+0.6, +1.7]
LaBSE	S_3	[0.66, 0.72]	[0.55, 0.67]	[0.47, 0.60]	[-4.5, +3.7]	[-0.4, +0.6]	[-0.6, +0.3]
	S_4	[0.63, 0.67]	[0.54, 0.61]	[0.55, 0.62]	[+3.0, +8.9]	[-1.0, +0.2]	[-1.2, +0.3]
	S_5	[0.61, 0.69]	[0.52, 0.66]	[0.48, 0.62]	[-2.9, +9.6]	[-1.3, +2.1]	[-2.0, +2.4]
	S_6	[0.67, 0.70]	[0.54, 0.60]	[0.61, 0.68]	[+13.0, +18.5]	[-3.8, -1.3]	[+0.2, +1.3]
MARBERTv2	S_3	[0.66, 0.72]	[0.59, 0.69]	[0.55, 0.66]	[-2.2, +5.3]	[-0.4, +0.6]	[-1.2, +0.5]
	S_4	[0.69, 0.73]	[0.58, 0.65]	[0.61, 0.68]	[+10.0, +15.7]	[-0.1, +0.9]	[+1.6, +3.9]
	S_5	[0.70, 0.77]	[0.55, 0.69]	[0.47, 0.62]	[+7.0, +19.8]	[-1.9, +1.2]	[-1.3, +2.9]
	S_6	[0.69, 0.72]	[0.52, 0.58]	[0.59, 0.66]	[+1.7, +7.1]	[-0.9, +0.3]	[+0.3, +1.8]

Table 10: 95% intervals for the mechanism measurements of Table 3 (1,000 bootstrap resamples of test sentences, seed-averaged within each resample; estimators not refitted).

	views	reference headroom (two resampled classifiers)				
Enc.	headroom	bootstrap	80% sub.	50% sub.	boot., $3C$	boot., $C/3$
mE5	8.0–14.1	6.1–9.1	2.9–4.0	4.8–6.8	8.0–10.9	4.2–7.3
XLM-R	7.3–10.7	3.4–5.1	2.1–3.0	1.7–3.6	5.1–7.2	2.0–4.2
LaBSE	7.9–14.9	6.6–7.4	3.3–3.9	6.1–7.0	7.2–8.6	5.4–6.2
MARBERTv2	11.5–16.5	10.6–15.3	4.7–6.3	9.4–14.5	14.8–21.9	8.2–11.3

Table 11: Sensitivity of the oracle reference (range over the four sets). "Views headroom": oracle over x and $T(x)$ minus the best view; other columns: the same for two classifiers differing only in their training sample (bootstrap as in Table 3, 80%/50% subsamples, or bootstrap with $C \times 3$ or $C/3$), applied to x; mean of 10 pairs.

with seed $1000 + s$; the 20% part tunes thresholds.

Few-label audit. `check_fewlabel.py` replays every targeted and random label draw of all eight encoder–regime runs (8,000 draws), saves the sampled source identifiers, and counts any sampled label from the scored fold, any shared source identifier or normalized text between sampled and scored sentences, and any targeted sentence that is not a disagreement: none. A rerun of `adapt.py` for mE5 with index logging matches the replay in all 1,000 draws and the archived results in all 2,500 rows. In 375 targeted draws (S_5, k=128) fewer than k disagreements were available.

Per-example exports. `results/export/per_example` gives, per encoder, regime and set, sentence and source identifiers, labels, both views' probabilities and every system's λ=0 prediction per seed. `scripts/reproduce_stats.py` (NumPy and pandas only) recomputes from them every difference, interval and p-value of Tables 2 and 7 for all four encoders exactly.

Mean vs input-dependent cost (exploratory). Since $T(x)$ is known at the decision point, the token model also gives each input's own avoidable cost $c_1(x)$. Replacing $\lambda\bar{c}$ by $\lambda c_1(x)$ ($\lambda > 0$, all encoders, sets, seeds) lowers cost by 0–0.11 passes and changes macro-F1 by -2.0 to $+0.6$ (means -0.03 and -0.13); it is no worse on both in 29 of 80 cells and trades accuracy for cost in 48: it moves along the frontier.

Costs. c_T is the measured normalizer time (about 0.4% of a pass). c_0, c_1 are *estimated* per sentence by a per-encoder model $a + b$·tokens fitted to measured batch-1 latencies of 300 sentences per set and view (e.g. mE5 $a = 19.6$ ms, $b = 0.32$ ms/token, $R^2 = 0.68$), as used in Figure 2 and Table 9. `measured_cost.py` runs the deployed pipeline on raw text at batch size 1 with nothing cached (pinned encoder, the trained head refitted deterministically, which reproduces the cached probabilities exactly, normalizer, features, each seed's router and λ=0 threshold, second pass only when selected). On all 9,220 test sentences, none of the 184,400 live routing decisions (4 encoders $\times$ 5 seeds) or final predictions differs from the archived ones (probabilities within $1.6 \cdot 10^{-4}$). Timed back to back on 200 sentences per set: B0 22–31 ms; relative costs within -0.02 to $+0.05$ of the estimates for B1 and R, $+0.03$ to $+0.07$ for SelectNorm (features are not in the token model). mE5 was re-timed after a concurrent job overlapped it (both runs archived).